HR Employee Attrition Analysis

Project report by Drishti Kori · Tools: Excel (PivotTables), Tableau · October 2026

1. Executive Summary

This project analyses attrition across a workforce of **1,470 employees**, of whom **237 (16.1%) have left**. The goal was to find out where attrition is concentrated and which factors are most closely associated with it, so that HR can target retention efforts instead of spreading them thinly.

The headline findings:

- **Overtime is the strongest single signal.** Employees working overtime leave at 30.5%, nearly three times the 10.4% rate of those who do not. They are 28% of the workforce but account for 54% of all leavers.
- **Attrition is concentrated in early-career, low-paid, short-tenure staff.** Employees aged 18–24 (39.18%), in their first year (34.88%) or in the Low Income band (28.6%) are the highest-risk groups.
- **Sales Representatives are the hotspot role** at 39.76%, more than double the company average.
- **The risk compounds.** Low-income employees who work overtime leave at 56.1%.
- **Delayed promotion is not a driver in this data.** Attrition is slightly *lower* for employees with long gaps since their last promotion.

2. Objective and Scope

The project set out to answer ten business questions, each built as a PivotTable (PT01–PT10) and the main ones surfaced in a Tableau dashboard:

1. How large is the workforce and what share has left?
2. Which departments lose the most people?
3. Which job roles have the highest attrition?
4. Are employees who work overtime more likely to leave?
5. Which age groups leave most?
6. Is lower job satisfaction linked to higher attrition?
7. Are newer employees more likely to leave?
8. How does attrition vary by income band?
9. Does time since last promotion matter?
10. Do employees with poorer work-life balance leave more?

Attrition rate throughout is the number of employees who left divided by the number of employees in the group.

3. Data and Preparation

The raw file contained 1,482 rows and 35 columns covering demographics, role and department, pay, tenure, satisfaction scores, overtime and attrition status.

Cleaning steps

- Removed 12 blank or incomplete rows, leaving **1,470 clean records** with no duplicate employee numbers.
- Ran a missing-value check across all 35 columns on the Data_Quality sheet.
- Dropped columns with no analytical value (a constant employee count, standard hours and the over-18 flag).
- Renamed fields to a consistent format (for example `Job_Role`, `Monthly_Income`).

Derived fields

- `Attrition_Flag`: 1 if the employee left, 0 otherwise, so averages give attrition rates directly.
- `Age_Group`: 18–24, 25–34, 35–44, 45–54, 55–60.
- `Tenure_Group`: 0–1, 2–4, 5–9, 10–20 and 20+ years.
- `Income_Band`: Low (1,009–2,996), Medium (3,022–5,993), High (6,029–9,998) and Very High (10,008–19,999) monthly income.
- `Promotion_Delay_Band`: 0–1, 2–4, 5–9 and 10+ years since last promotion.
- `Work_Life_Balance_Band`: the 1–4 score relabelled Bad, Fair, Good and Excellent.

Data quality note. In the Clean_Data sheet, the `Business_Travel` column mixes two spellings of the same categories

(for example *Travel Frequently* and *Travel_Frequently*). I standardised these before analysing travel; the sheet itself should be corrected so that any pivot or dashboard built on it groups them together.

4. Methodology

I used descriptive analysis: grouping employees by one attribute at a time, calculating the attrition rate per group, and comparing it with the company average of 16.1%. PivotTables in Excel produced the ten core views, and a Tableau dashboard presents the main ones together with a KPI panel. Where a single factor looked important, I cross-checked it against a second one (for example overtime by income band) to see whether the effects overlap.

5. Key Findings

5.1 Overall attrition

Status	Employees	Share
Stayed	1,233	83.9%
Left	237	16.1%
Total	1,470	100%

5.2 Department and job role

Department	Employees	Attrition rate
Sales	446	20.6%
Human Resources	63	19.0%
Research & Development	961	13.8%

Sales has the highest rate among the large departments. The HR figure rests on only 63 people, so it should be read with caution.

The role view is sharper than the department view:

Job role	Attrition rate
Sales Representative	39.8%
Laboratory Technician	23.9%
Human Resources	23.1%
Sales Executive	17.5%
Research Scientist	16.1%
Manufacturing Director	6.9%
Healthcare Representative	6.9%
Manager	4.9%
Research Director	2.5%

Attrition falls steadily with seniority: the front-line roles at the top of the table lose the most people, while director and manager roles lose very few.

5.3 Overtime

Overtime	Employees	Leavers	Attrition rate
No	1,054	110	10.4%
Yes	416	127	30.5%

Overtime is the clearest driver in the dataset. The gap is especially large in the hotspot roles: among Sales Representatives, 67% of those working overtime left versus 29% of those who did not; among Laboratory Technicians the figures are 50% and 16%.

5.4 Age and tenure

Age group	Employees	Attrition rate
18–24	97	39.2%
25–34	554	20.2%
35–44	505	10.1%
45–54	245	10.2%
55–60	69	15.9%

Tenure	Attrition rate
0–1 years	34.9%
2–4 years	18.1%
5–9 years	11.1%
10–20 years	10.0%
20+ years	12.1%

Leavers are on average younger (33.6 versus 37.6 years). More than a third of employees in their first year leave, which points to onboarding and early-career experience as an area to examine. The small uptick in the 55–60 group is consistent with retirement and should not be read as dissatisfaction.

5.5 Income

Income band	Employees	Attrition rate
Low	395	28.6%
Medium	519	12.7%
High	275	12.0%
Very High	281	8.9%

The median monthly income of leavers is 3,202 compared with 5,204 for those who stayed. The sharpest break is between the Low band and everyone else, which suggests a pay floor effect more than a gradual relationship.

5.6 Satisfaction, work-life balance and promotion

Job satisfaction	Employees	Attrition rate
1 (lowest)	289	22.8%
2	280	16.4%
3	442	16.5%
4 (highest)	459	11.3%

Lower job satisfaction goes with higher attrition, but the pattern is moderate rather than dramatic. Work-life balance shows a stronger effect at the bottom end: employees rating it 1 leave at 31.3% (80 people), against 14.2% for those rating it 3.

Promotion delay shows no useful relationship: attrition is 17.0% for those promoted within the last year and 12.5% for those waiting ten or more years. In this data, time since promotion does not explain who leaves.

5.7 Additional patterns

A few further cuts supported the main story:

- **Marital status:** single employees leave at 25.5%, against 12.5% for married and 10.1% for divorced employees.
- **Business travel:** frequent travellers leave at 24.9%, rarely-travelling at 15.0% and non-travellers at 8.0%.
- **Stock options:** employees with no stock options leave at 24.4%, versus 9.4% and 7.6% at option levels 1 and 2.
- **Job level:** level 1 staff leave at 26.3%, against 9.7% at level 2.
- **Distance from home:** leavers live slightly further away on average (10.6 versus 8.9).
- **Gender:** the gap is small (male 17.0%, female 14.8%).

5.8 Combined effect: overtime and income

Income band	No overtime	Overtime
Low	17.4%	56.1%
Medium	8.5%	24.1%
High	8.0%	22.4%
Very High	6.6%	14.1%

Overtime raises attrition in every income band, and its effect is largest where pay is lowest. More than half of low-income employees working overtime left, making this the highest-risk profile found in the project.

6. Recommendations

1. **Manage overtime first.** Review workload and staffing in the roles where overtime and attrition coincide, starting with Sales Representatives and Laboratory Technicians. Even a partial reduction in overtime targets the factor most closely tied to leaving.
2. **Review pay at the bottom of the scale.** The Low Income band has more than double the attrition of the next band. A targeted pay review or progression path for entry-level roles is likely to be more effective than across-the-board increases.
3. **Strengthen the first year.** With 34.9% of first-year employees leaving, invest in onboarding, mentoring and early check-ins, especially for under-25s.
4. **Investigate Sales Representative turnover specifically.** At nearly 40%, this role needs its own review of targets, pay structure and management support.
5. **Extend equity-based retention.** Employees with no stock options leave far more often; consider whether options or similar incentives can reach more junior staff.
6. **Track work-life balance and satisfaction in pulse surveys,** and follow up on the lowest scores, particularly work-life balance ratings of 1.
7. **Do not prioritise promotion speed** as a retention lever on this evidence; other factors matter more.

7. Limitations

- **Association, not causation.** The analysis shows which groups leave more often, not why. Overtime, for example, may be a symptom of understaffing as much as a cause of leaving.
- **One factor at a time.** Most views are single-variable. Factors overlap (young employees are often junior, low-paid and in front-line roles), so the effects cannot be fully separated without a multivariate model.
- **No statistical testing.** Differences were compared descriptively; no significance tests or confidence intervals were applied.
- **Small groups.** Some segments, such as Human Resources (63 employees) and work-life balance rating 1 (80 employees), are small, so their rates are less stable.
- **Snapshot data.** There is no leaving date, so trends over time and the timing of departures cannot be analysed.

8. Next Steps

- Fix the inconsistent `Business_Travel` labels in the clean dataset and extend the Data_Quality sheet to cover duplicates and category consistency, not only missing values.
- Build a logistic regression or random forest model to estimate each factor's independent effect and to produce an attrition-risk score for current employees.
- Add overtime-by-role and overtime-by-income views to the Tableau dashboard, since the combined effect is the most actionable result.

9. Deliverables

- **Excel workbook:** Raw_Data, Clean_Data, Data_Quality and Pivot_Tables sheets (PT01–PT10).
- **Tableau dashboard (HR Attrition Analysis Dashboard):** KPIs and attrition by department, job role, overtime status, age group and tenure group.